

2026 DEFRA, local council and Urban Observatory sensor, data analysis

Following a meeting with council member, Michael Terry, we were made aware of NCC's restrictions in the use of data from calibrated DEFRA approved and owned AURN instrumentation and locally-managed automatic monitoring sensors. The Urban Observatory (UO) has a collection of precision MONITOR sensors and MESH sensors. The UO MESH sensors are not calibrated and not approved by DEFRA and therefore NCC cannot use them, the UO MONITOR sensors are in need of verification before they can be used for any further work. We begin by looking at a thorough data analysis of the UO MESH and MONITOR sensors against the official, calibrated DEFRA approved and owned AURN instrumentation and locally-managed automatic monitoring sensors. Any and all analysis has been conducted on 2026 data.

For the purposes of this report we will use the following acronyms:

- DEFRA owned and approved AURN sensors (DEFRA)
- locally managed automatic monitoring sensors (Local)
- UO precision MONITOR sensors (UO-Mon)
- UO MESH sensors (UO-Mesh)

Summary

We have highlighted the following as areas for priority action:

- For PM2.5 data, the UO-Mon and UO-Mesh sensors have low agreement. We will proceed with using the precision sensors only.
- Some UO-Mesh and UO-Mon sensors have faulty wind direction and speed data, shown as constant over the course of a day, week and month, we have taken some decommissioned sensor data from 2025 to produce analysis.
- DEFRA Newcastle Centre sensor is systematically rounding PM2.5 values to nearest integer, this will be brought to NCC attention.
- To validate the UO-Mon sensors we compared them to their closest DEFRA/Local sensors, most proved good agreement, some had a strong exponential relationship which requires some further analysis. Some of the disagreement can be attributed to distance between sensors, however this was not a consistent factor. We would appreciate expert UO opinion here to explore other extenuating factors for the purpose of UO-Mon sensors validation.

This report is structured as follows...

Explain why look at monthly and weekly data, and not just monthly data.

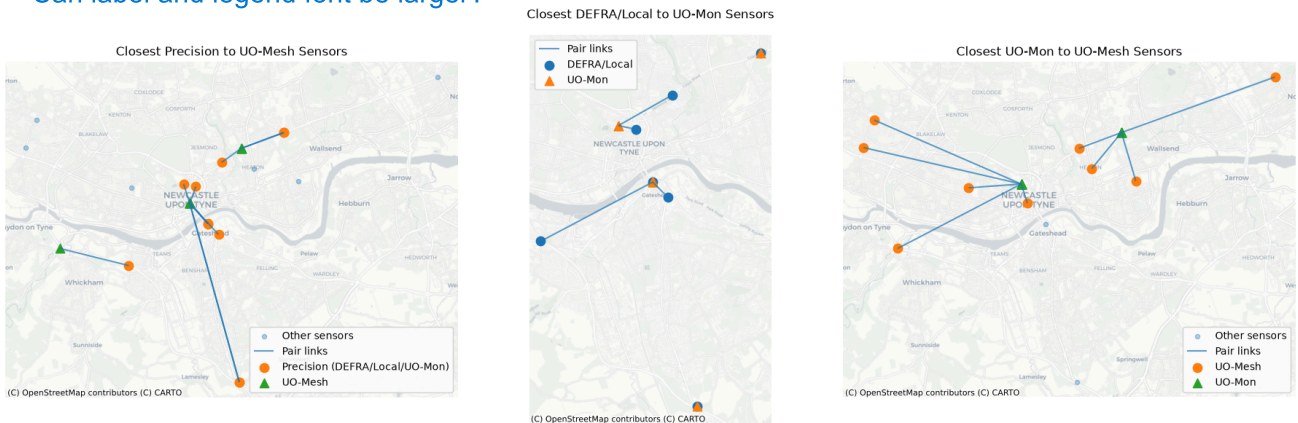
Maps

The three maps in Figure 1 show closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh

sensors.

Figure 1

Can label and legend font be larger?



Monthly data

We took March 2026 to look at correlation comparisons between the closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest **Defra**/Local sensors to UO-Mon sensors. March was chosen as the dataset had reliable consistent data, i.e. minimal missing values and missing sensor data.

PM2.5 comparisons between close sensors

Closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors, Figure 2 below shows scatter plots of these relationships. These show little to no reliability of the UO mesh sensors.

Figure 2: comparison between UO-Mesh and closest precision sensors.

Make sure all plots/tables have captions and are on the same page with corresponding plots/tables.

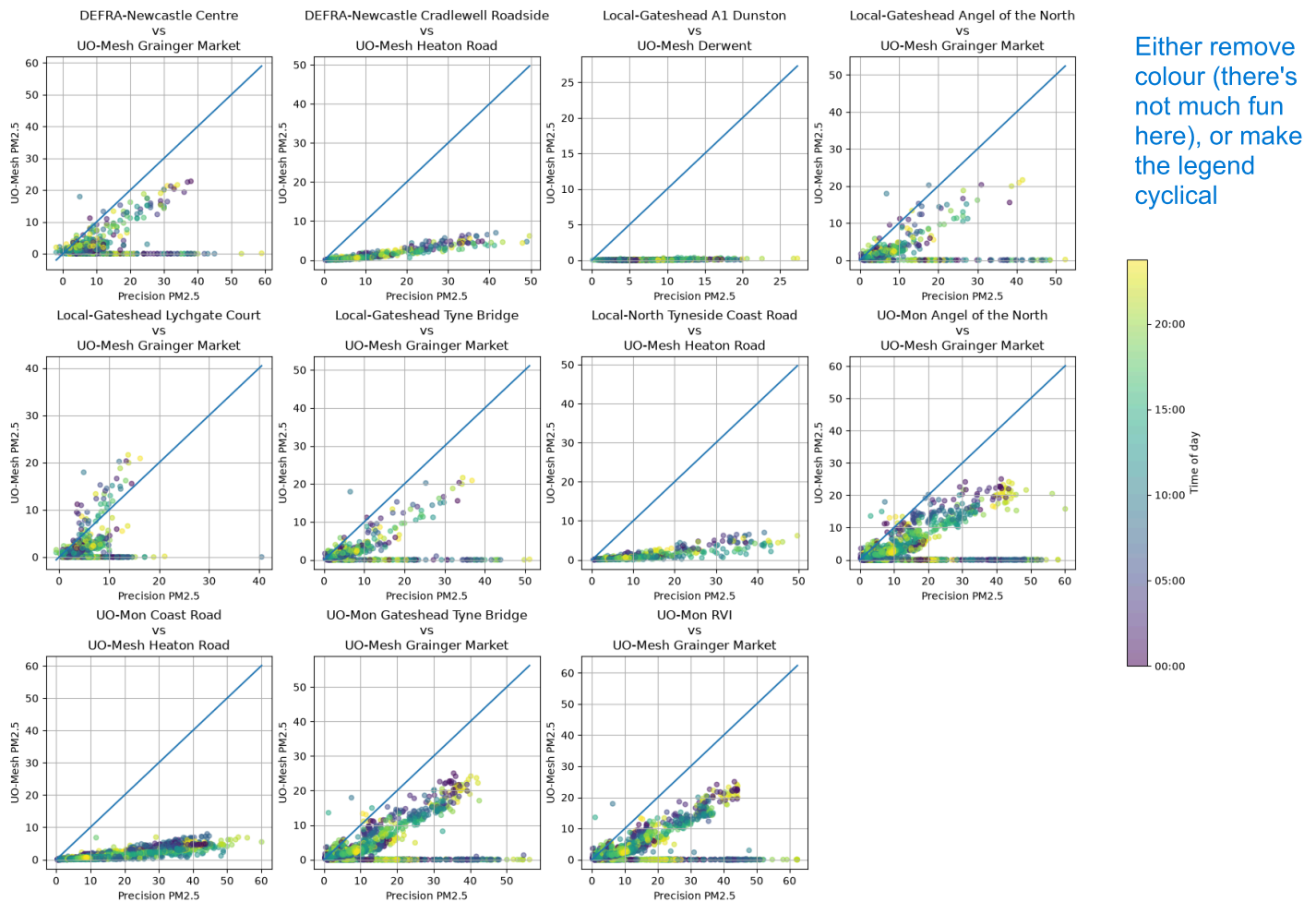


Figure 3 shows closest DEFRA/Local sensors to UO-Mon sensors, scatter plots and time series. This hopes to validate the UO-Mon sensor data and from both sets of plots we see good agreement between most sensors. A summary table of correlational information and distances between sensors is given in Table 1 from which we see high correlation between all close sensors. The scaling differences in the Loal A1 Dunston sensor and Local Gateshead Lychgate court sensors could be attributed to their further distance from thier closest UO-Mon if it were not for DEFRA Newcastle Cradlewell Roadside case which is further away from it's closest UO-Mon than the Lychgate sensor. It could also be attributed to the fact that we are comparing the same monitor sensor for 4 of the DEFRA/Local sensors as there are no closer UO-Mon sensors in the area. Another issue to raise is the case of DEFRA Newcastle Centre sensor, this is systematically rounding values to the closest integer, we will see this more clearly when analysing weekly data but here you can spot it in the time series plot.

Figure 3 include caption

The pairs of closest sensors (Angel/Coastroad/Tyne bridge) show very good agreement. One plot should only be UO-Mon with a closest non-UO sensor, and another plot can include relationships with other sensors e.g. (UO-Mon Tyne Bridge vs Local Dunston), but for UO-Mon validation this is not as relevant. We can show that as distance increases the agreement changes, which is to be expected. NOTE: in those plots with inconsistencies, can you double-check what happened around 19th/20th of March. Is there missing/interpolated data?

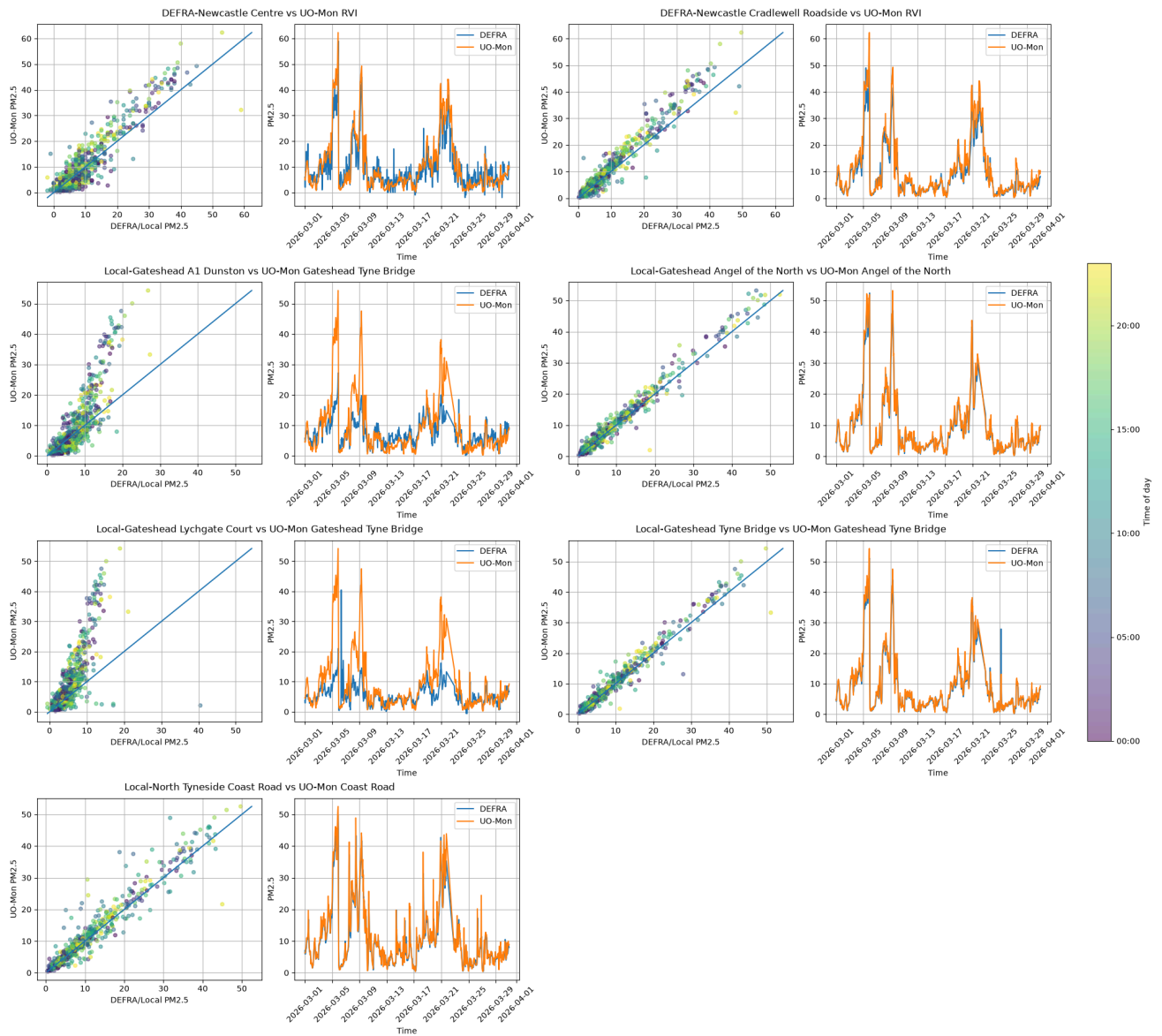
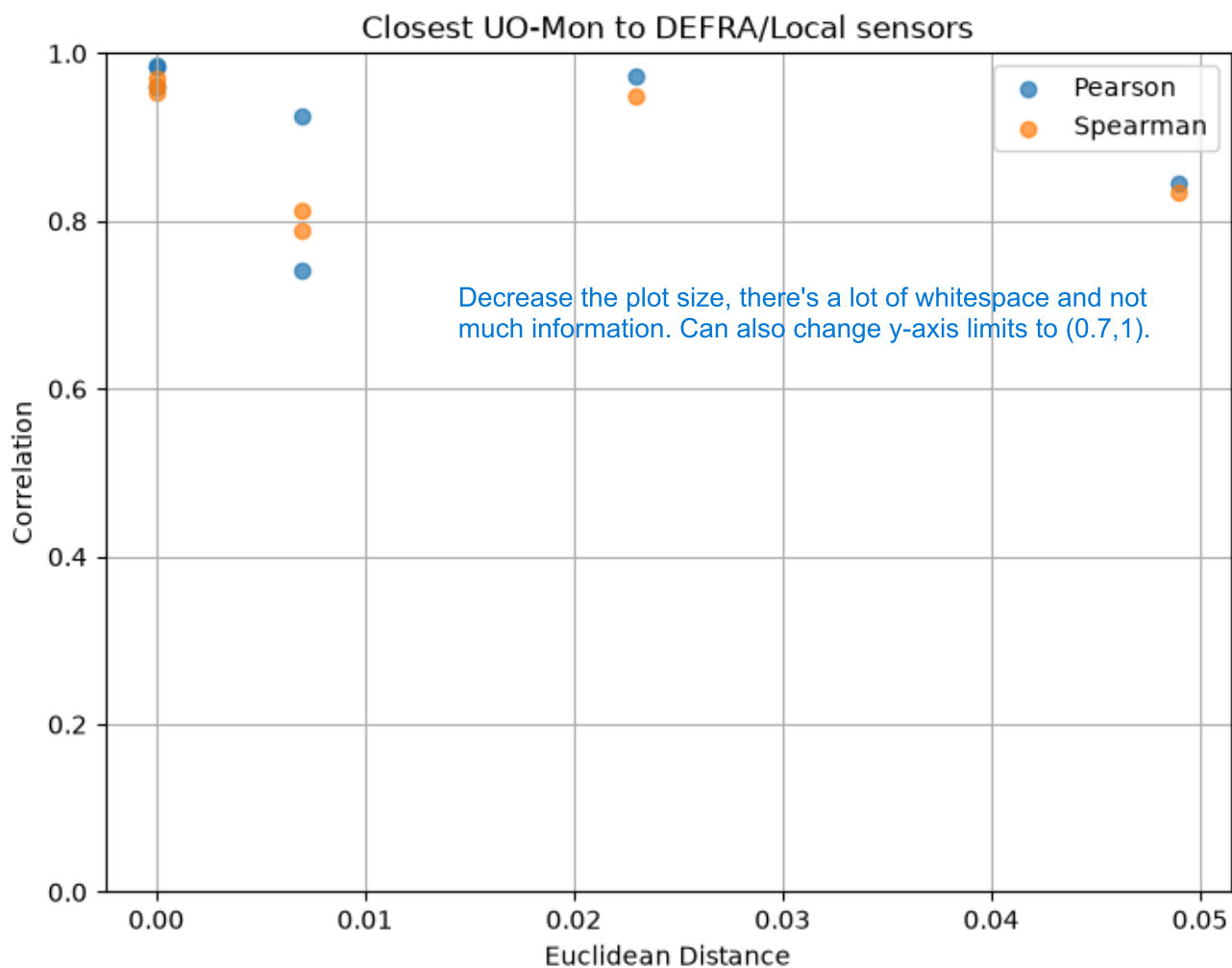


Table 1 caption, same page as table

	DEFRA/Local	UO-Mon	Distance km	Distance Euclidean	Pearsons corr	Spearman's corr
3	Local-Gateshead Angel of the North	UO-Mon Angel of the North	0.003	0.000	0.986	0.962
5	Local-Gateshead Tyne Bridge	UO-Mon Gateshead Tyne Bridge	0.006	0.000	0.983	0.971
1	DEFRA-Newcastle Cradlewell Roadside	UO-Mon RVI	1.643	0.023	0.974	0.950
6	Local-North Tyneside Coast Road	UO-Mon Coast Road	0.002	0.000	0.961	0.954
0	DEFRA-Newcastle Centre	UO-Mon RVI	0.468	0.007	0.926	0.812
2	Local-Gateshead A1 Dunston	UO-Mon Gateshead Tyne Bridge	3.370	0.049	0.846	0.834
4	Local-Gateshead Lychgate Court	UO-Mon Gateshead Tyne Bridge	0.572	0.007	0.741	0.789



Not everyone will know what this means, just use Distance (km)

Weekly data

We took the last week (23rd-29th) of March 2026 to look at correlation comparisons in PM2.5 concentrations between the UO-Mesh sensors to UO-Mon sensors. We also looked at correlational comparisons of PM2.5 and NOx/NO2 concentrations in the same sensors for all DEFRA, Local and UO-Mon sensors.

PM2.5 comparisons between close sensors

Figure 4 shows closest UO-Mon sensors to UO-Mesh sensors scatter plots. These show little to no agreement of the UO-Mesh sensors even with the UO-Mon sensors. A summary table of their correlational information and distances is given in Table 2.

Can you comment on the summary from the table below? Is there anything interesting? If you included it, need to say something about it. Alternatively, can include this information as boxes in top left corners of every plot e.g. vector("dist=0.5 km", "Pears=0.3", "Spear=0.2")

Figure 4

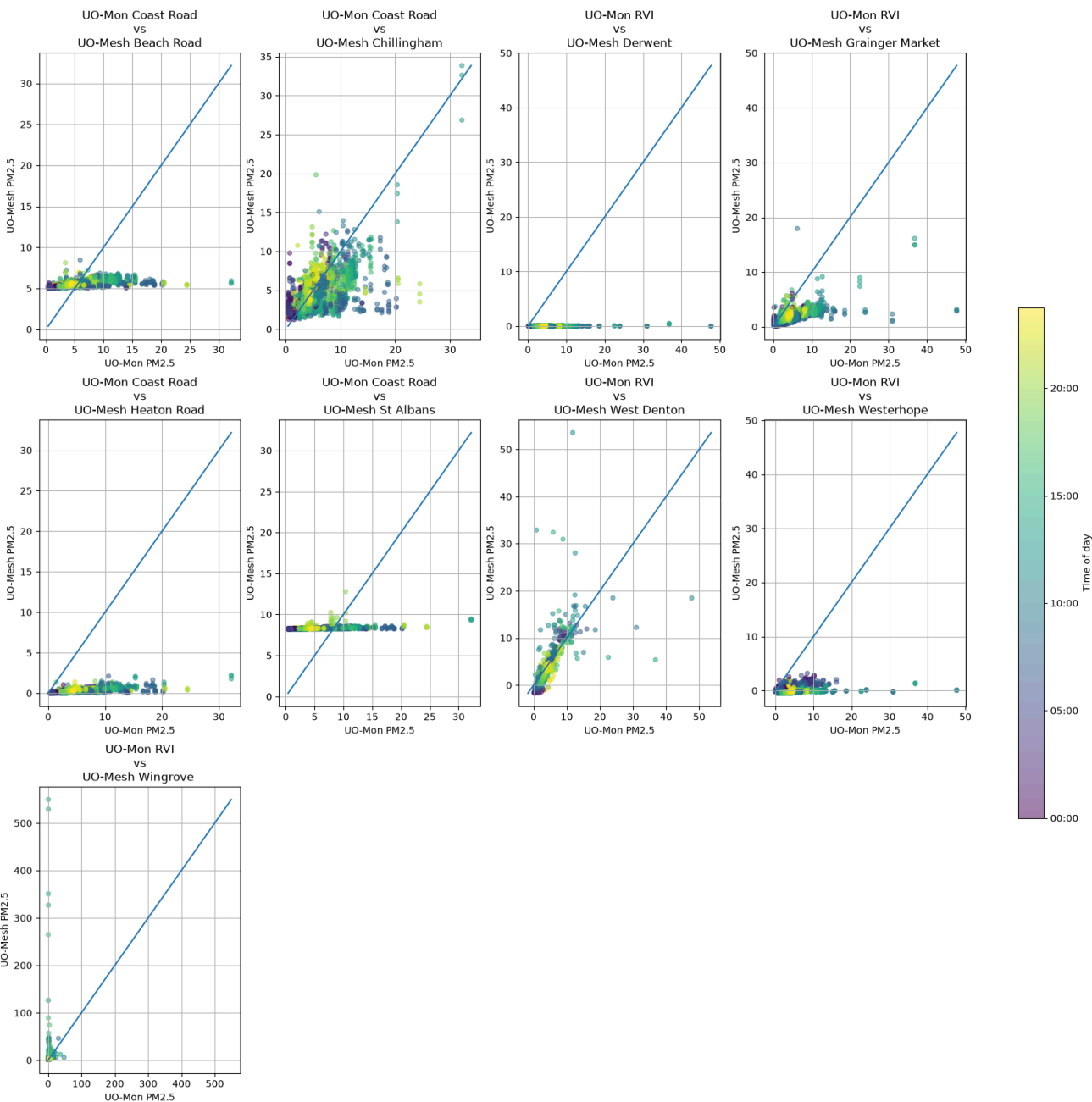


Table 2

	UO-Mesh	UO-Mon	Distance km	Distance Euclidean	Pearsons corr	Spearman's corr
4	UO-Mesh Heaton Road	UO-Mon Coast Road	1.709	0.026	0.660	0.704
6	UO-Mesh West Denton	UO-Mon RVI	6.098	0.094	0.659	0.906
3	UO-Mesh Grainger Market	UO-Mon RVI	0.754	0.007	0.636	0.757
1	UO-Mesh Chillingham	UO-Mon Coast Road	1.757	0.021	0.579	0.611
0	UO-Mesh Beach Road	UO-Mon Coast Road	6.150	0.092	0.514	0.643
5	UO-Mesh St Albans	UO-Mon Coast Road	1.904	0.018	0.348	0.599
2	UO-Mesh Derwent	UO-Mon RVI	5.252	0.076	0.342	0.286
7	UO-Mesh Westerhope	UO-Mon RVI	6.045	0.089	0.220	0.241
8	UO-Mesh Wingrove	UO-Mon RVI	1.985	0.031	-0.058	0.638

NOx and NO2 comparisons between same sensors

First NOx and NO2 concentration comparison in the DEFRA sensors, the aim of this analysis is to discern if PM2.5 has any traffic dependency in the same way that NO2 and NOx behave. In Figure 5 scatter plots, we see very clearly here the systematic categorisation of the DEFRA Newcastle Centre sensor which we can discern as unreliable. Low correlation values for all can be seen in Table 3.

Figure 5

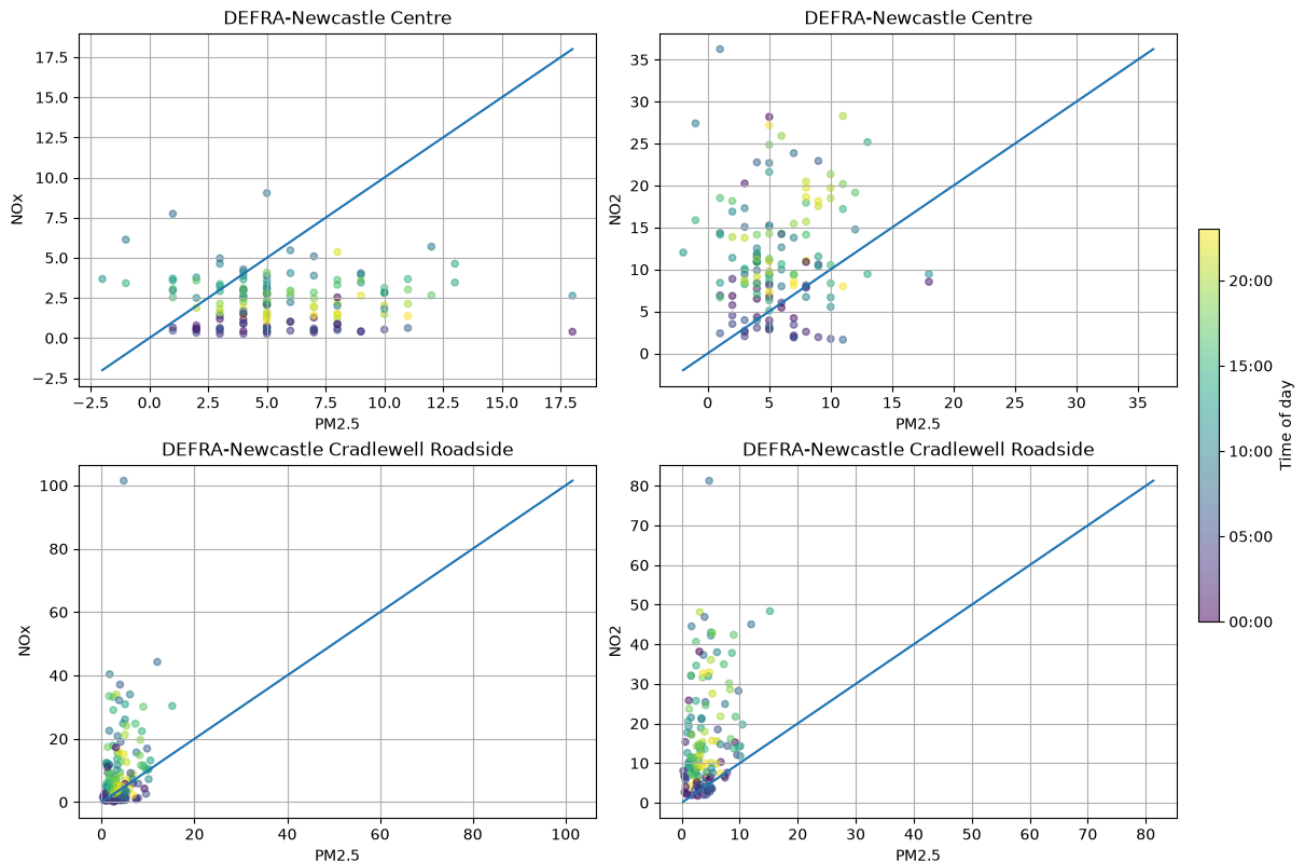


Table 3

	DEFRA	Pearsons corr NO _x	Spearman's corr NO _x	Pearsons corr NO ₂	Spearman's corr NO ₂
1	DEFRA-Newcastle Cradlewell Roadside	0.224	0.238	0.297	0.275
0	DEFRA-Newcastle Centre	-0.036	-0.008	0.059	0.074

In Figure 6, we seen scatter plots for NO₂ comparison with PM_{2.5} concentration for the Local sensors as they not not collect NO_x data. Low correlation for all can be seen in Table 4.

Figure 6

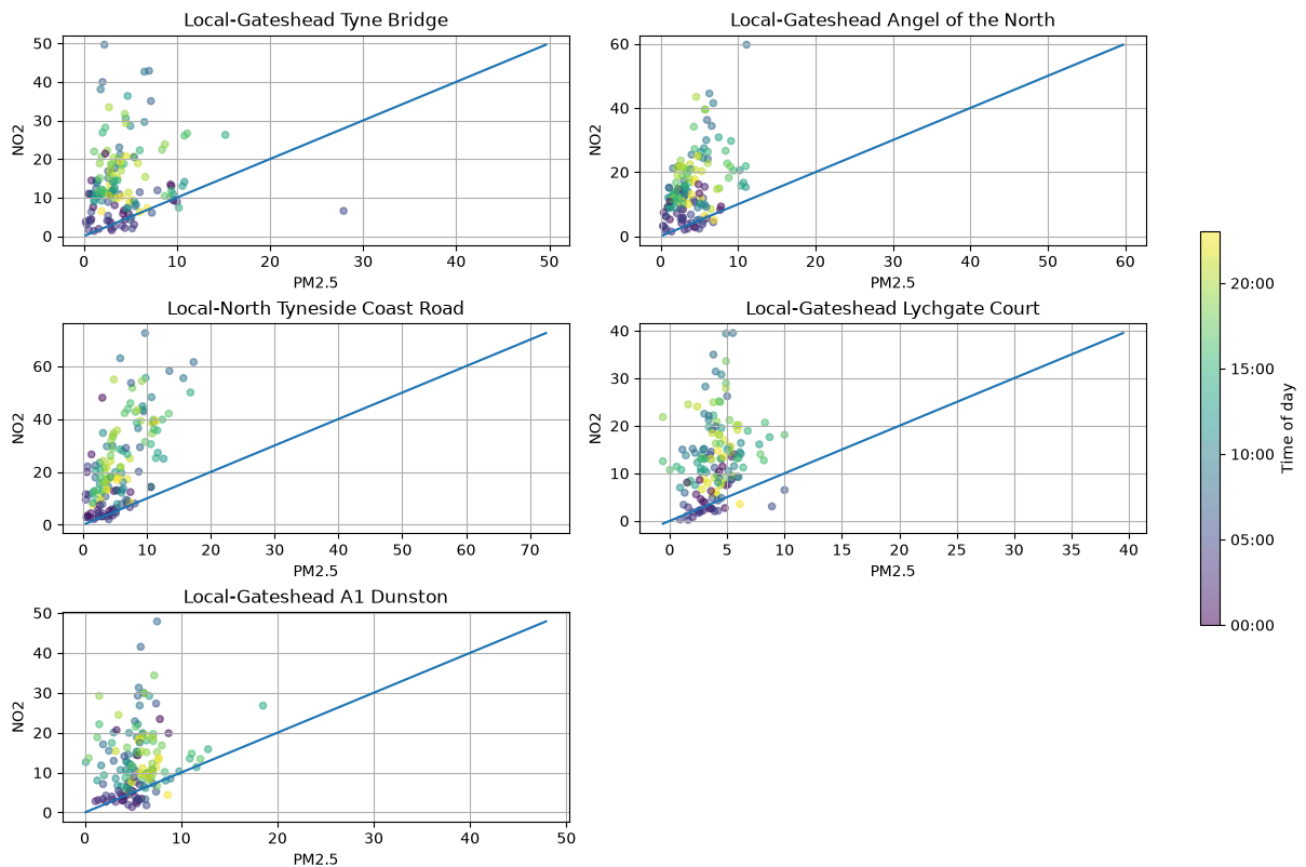


Table 4

	Local	Pearsons corr NO2	Spearman's corr NO2
2	Local-North Tyneside Coast Road	0.642	0.608
1	Local-Gateshead Angel of the North	0.426	0.384
3	Local-Gateshead Lychgate Court	0.217	0.312
4	Local-Gateshead A1 Dunston	0.210	0.247
0	Local-Gateshead Tyne Bridge	0.095	0.145

In Figure 7, scatter plots for NO_x comparison with PM_{2.5} concentration for the UO-Mon sensors for now. NO₂ concentration is still in process as downloading data seemed to be a problem in terms of data frequency. In Table 5, we see that some of the UO-Mon sensors show some positive correlation between PM_{2.5} and NO_x, however the other half show no correlation.

Figure 7

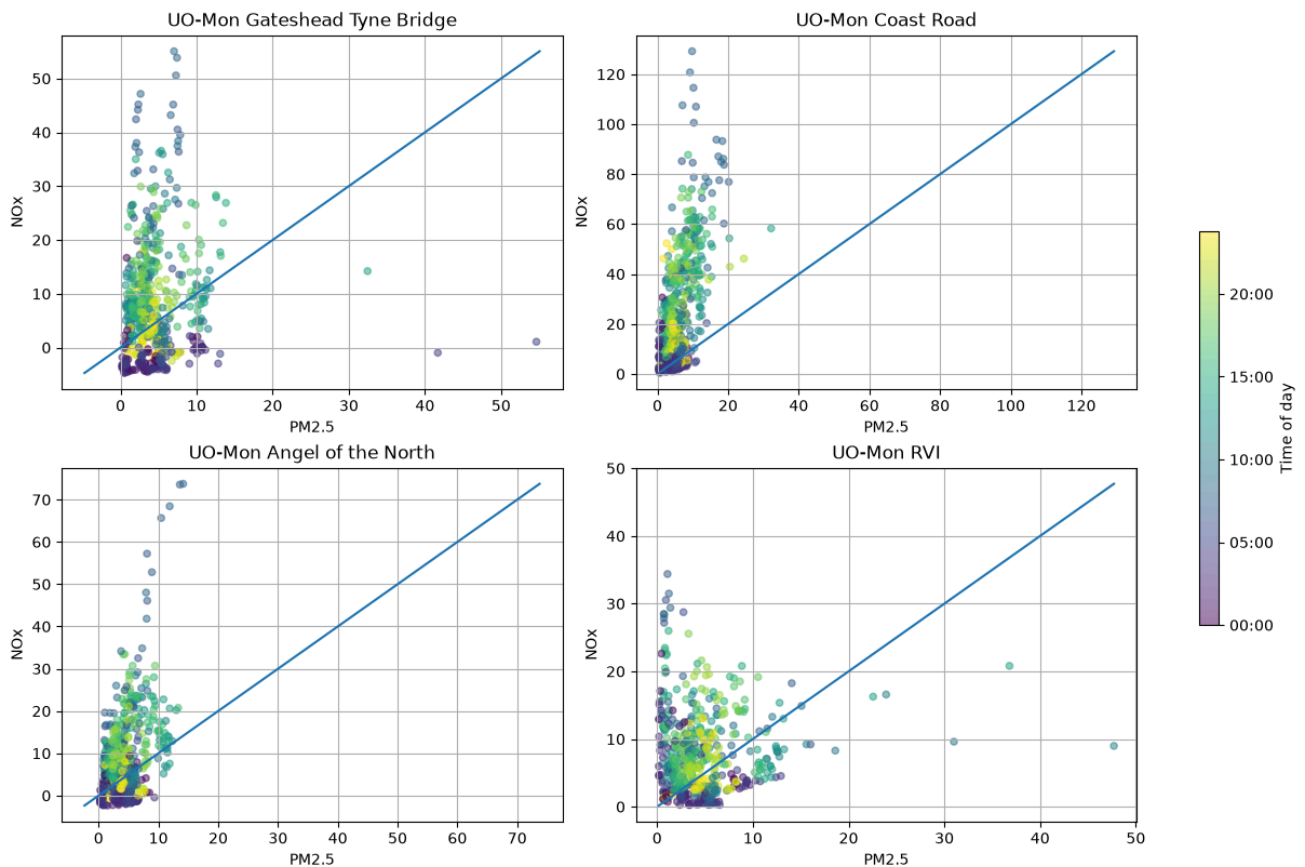


Table 5

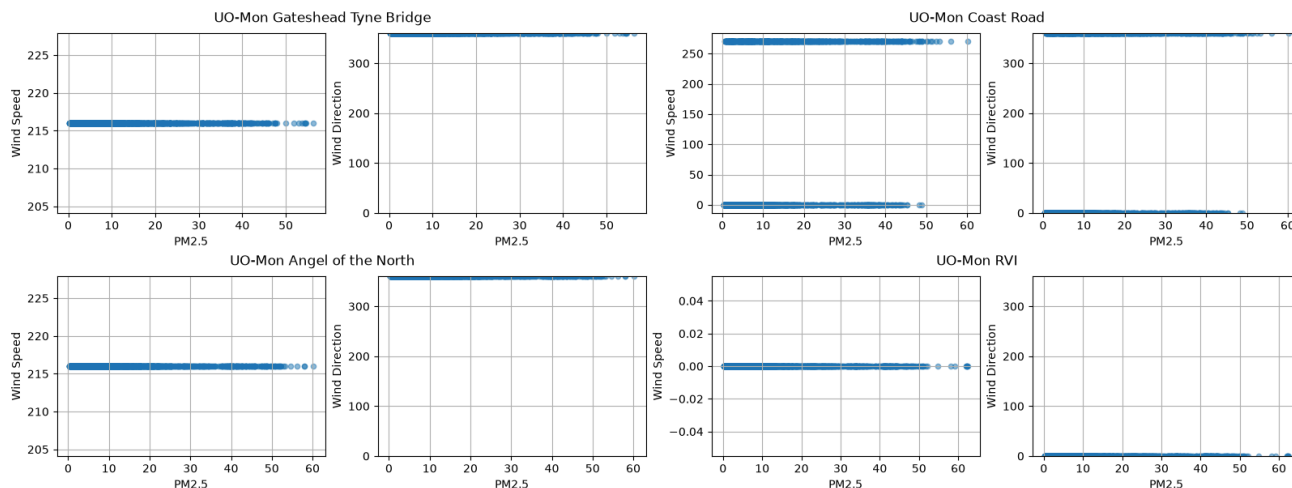
	UO-Mon	Pearsons corr NOx	Spearman's corr NOx
1	UO-Mon Coast Road	0.634	0.613
2	UO-Mon Angel of the North	0.446	0.339
0	UO-Mon Gateshead Tyne Bridge	0.074	0.123
3	UO-Mon RVI	0.035	-0.028

Attributes affecting PM2.5 concentraions

Wind direction and speed

Wind data from UO-Mon sensors was taken for the month of March 2026, which turned out to be faulty as seen from Figure 8 below showing constant speed and direction over the course of the entire day/week/month. Further to this we analysed wind data from two older UO sensors in Birtley and West Denton to see if we could dicern anything interesting.

Figure 8



The two UO sensors in Birtley and West Denton seem to have data throughout 2025, similarly to previous we looked at March 2025 data, we analysed how PM2.5 concentrations behaved in the closest UO-Mon sensors and Local sensors which seems to show a relation in wind direction. Figure 9 shows maps of the closest UO-Mon/Local sensors to the Birtley and West Denton sensors. Figure 10 shows scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest UO-Mon sensors and Figure 11 shows the same for closes Local sensors.

You have to say a bit more here (and I'm confused by the second column of plots in Figure 10). You can see that there is a relationship between PM2.5 concentration and wind direction, with low decrees of wind corresponding to high PM2.5 and around 180 degrees wind corresponding to lowest PM2.5...

Figure 9

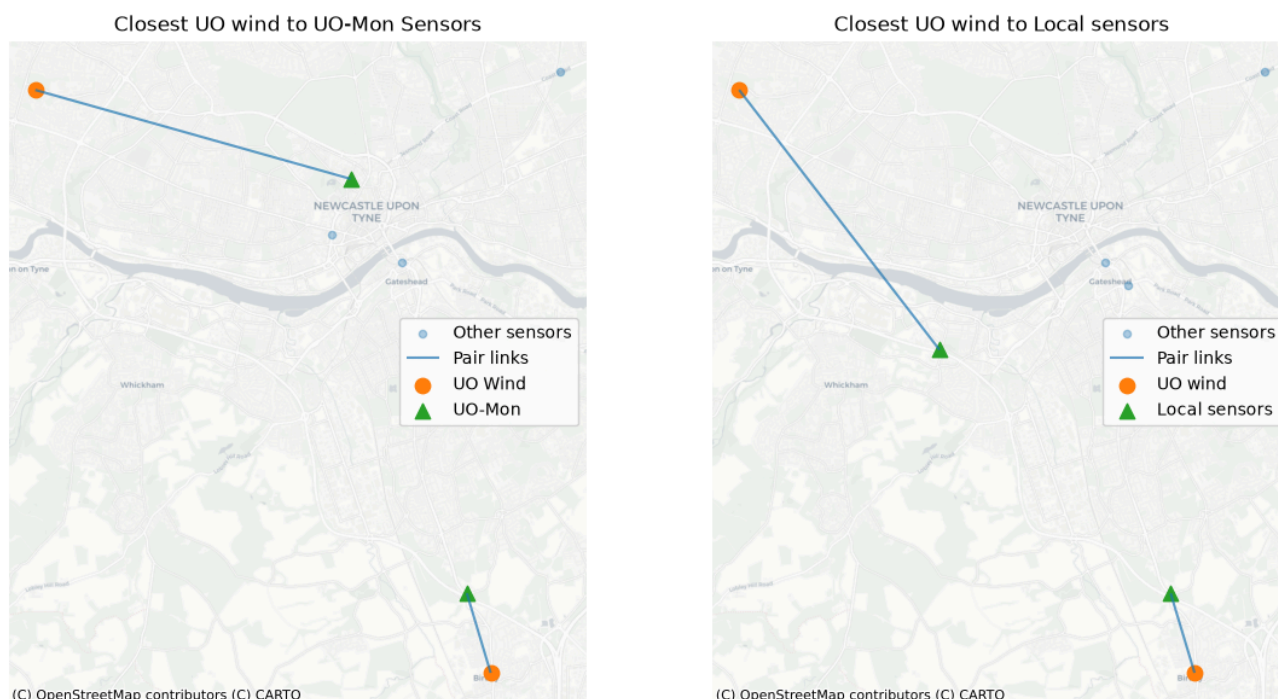


Figure 10

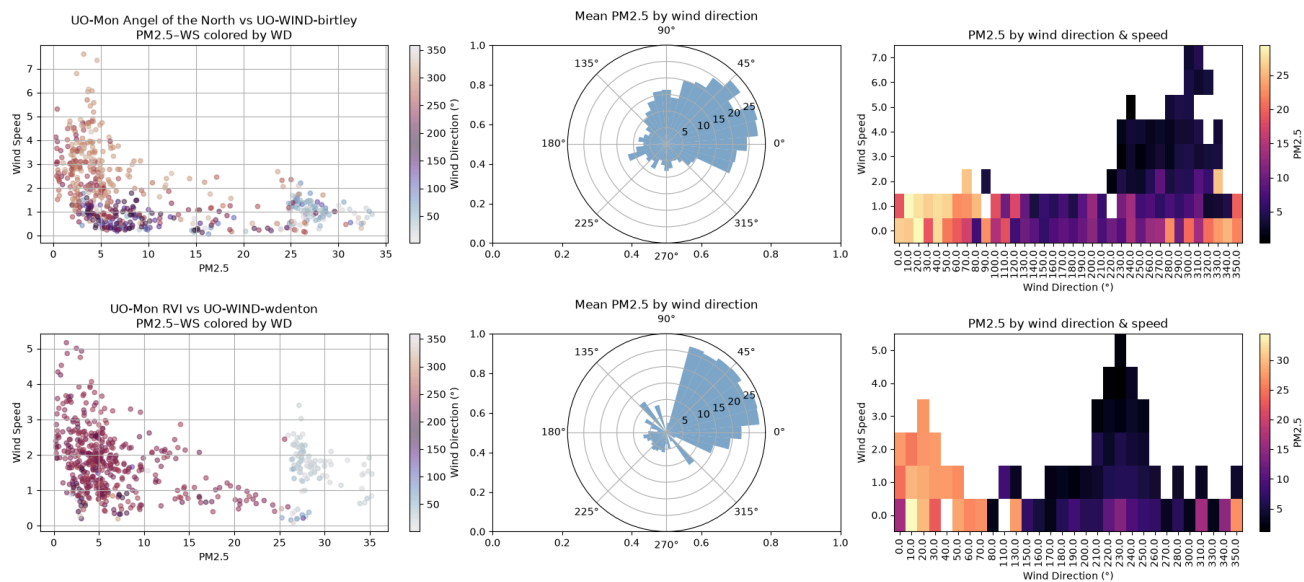
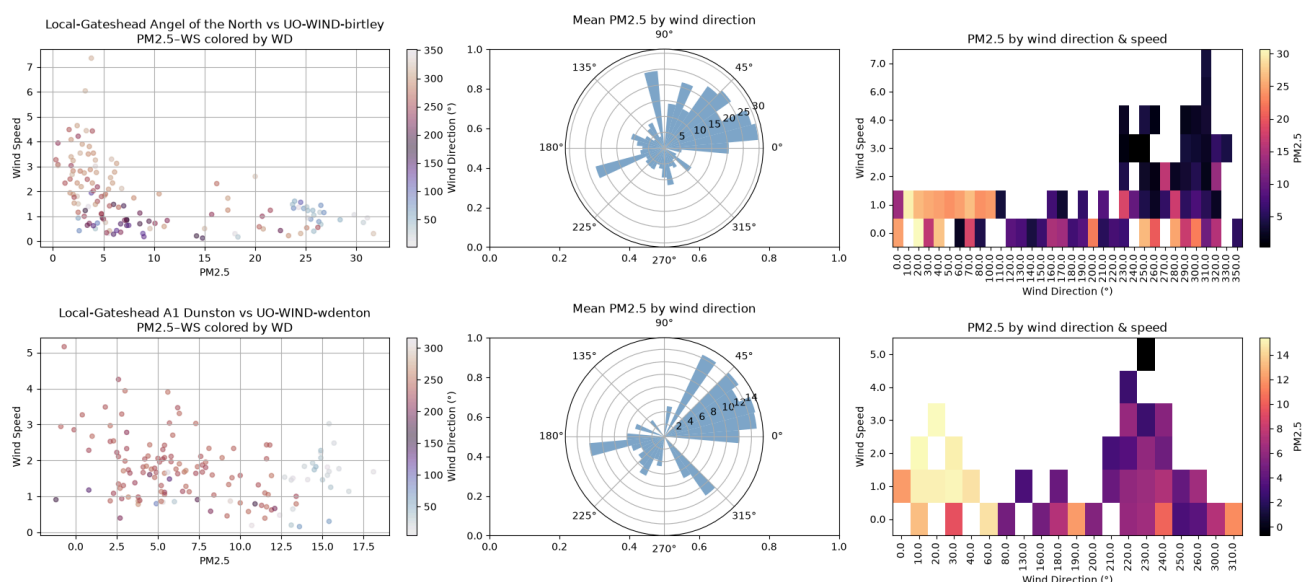


Figure 11



We are in the process of analysing other factors that may affect PM2.5 concentrations in certain locations such as temperature and altitude.